

Prepared for

Community Foundation for Muskegon County

Serving the Community Foundations of Muskegon, Mason, and Oceana Counties

January 27, 2020

Vanguard

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Agenda

I. Market recap

II. Portfolio performance and asset allocation

III. Economic and market outlook

Appendix

– Segregated accounts reporting

– Fund pages

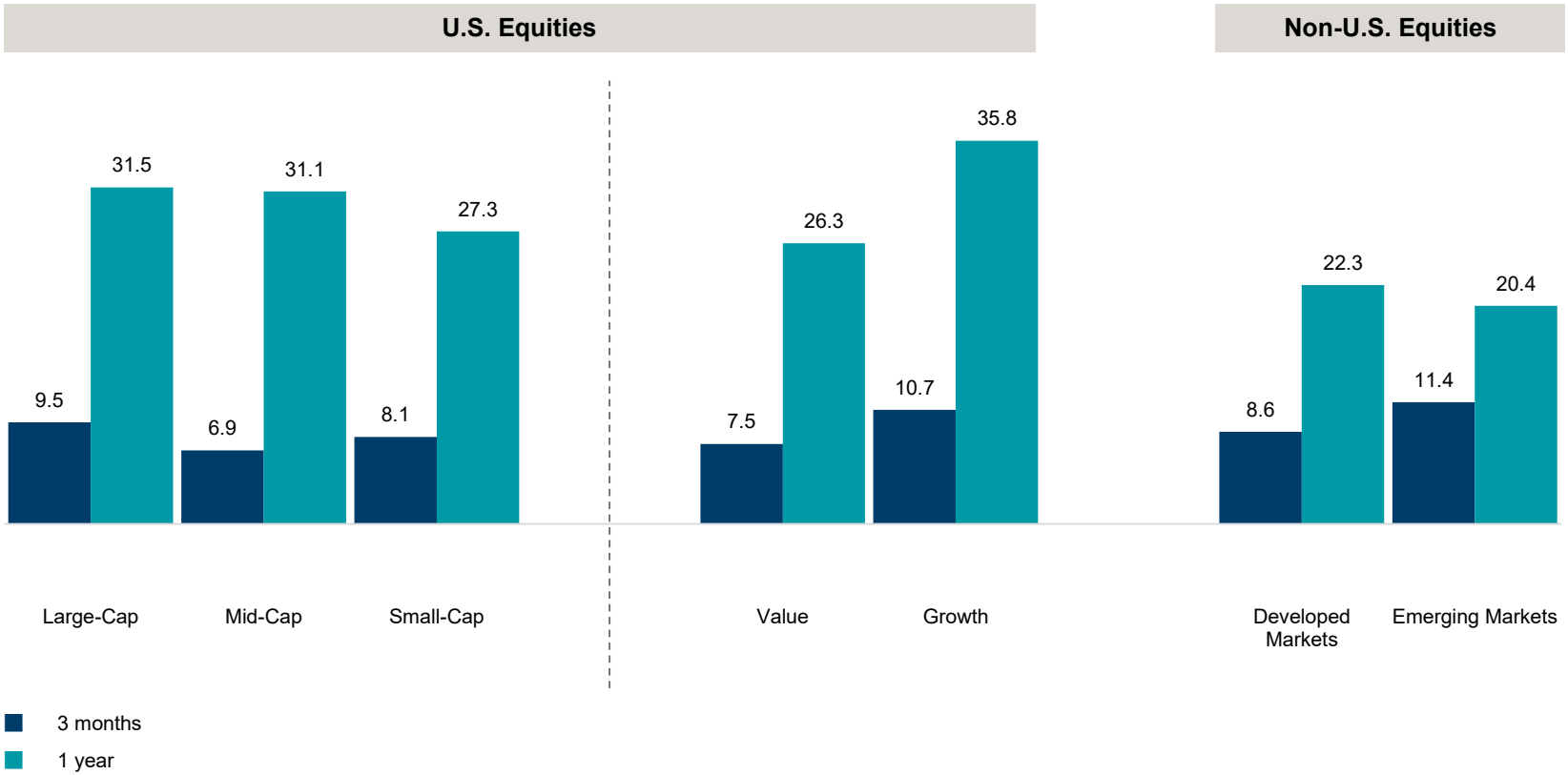
Presented by:

Matt Ruhl, CFA
Senior Investment Consultant
Vanguard Institutional Advisory Services®

Market recap

All segments of U.S. and Non-U.S. Equities had strong returns in 2019

Global equity market returns as of December 31, 2019 (%)

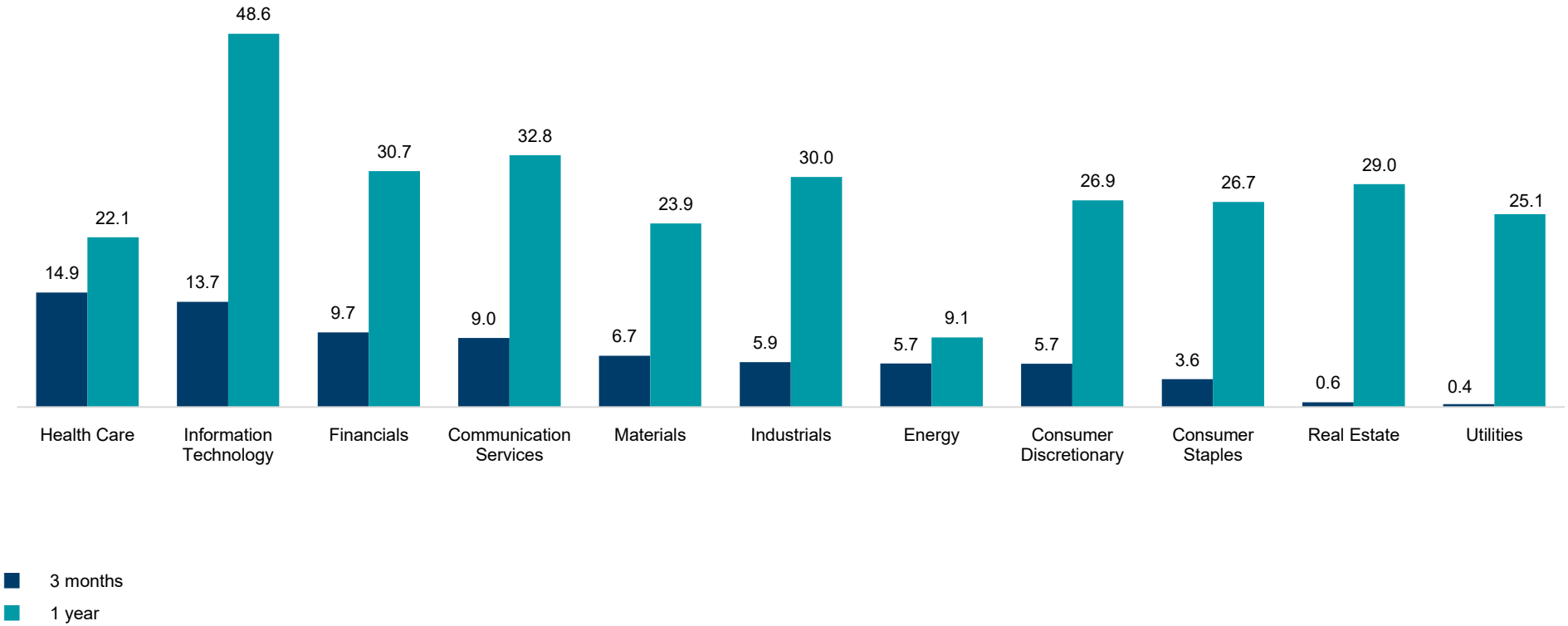


Past performance is no guarantee of future returns. The performance of an index is not an exact representation of any particular investment, as you cannot invest directly in an index.

Sources: FTSE, MSCI, Russell, CRSP and Dow Jones.
Large-cap (CRSP US Mega Cap Index), Mid-cap (CRSP US Mid Cap Index), Small-cap (CRSP US Small Cap Index); Value (Russell 3000 Value Index), Growth (Russell 3000 Growth Index); Developed markets (FTSE Developed All Cap ex-US Index), Emerging markets (FTSE Emerging Markets All Cap China A Inclusion Index).

All sectors have exhibited positive performance over the course of 2019

U.S. equity sector returns as of December 31, 2019 (%)

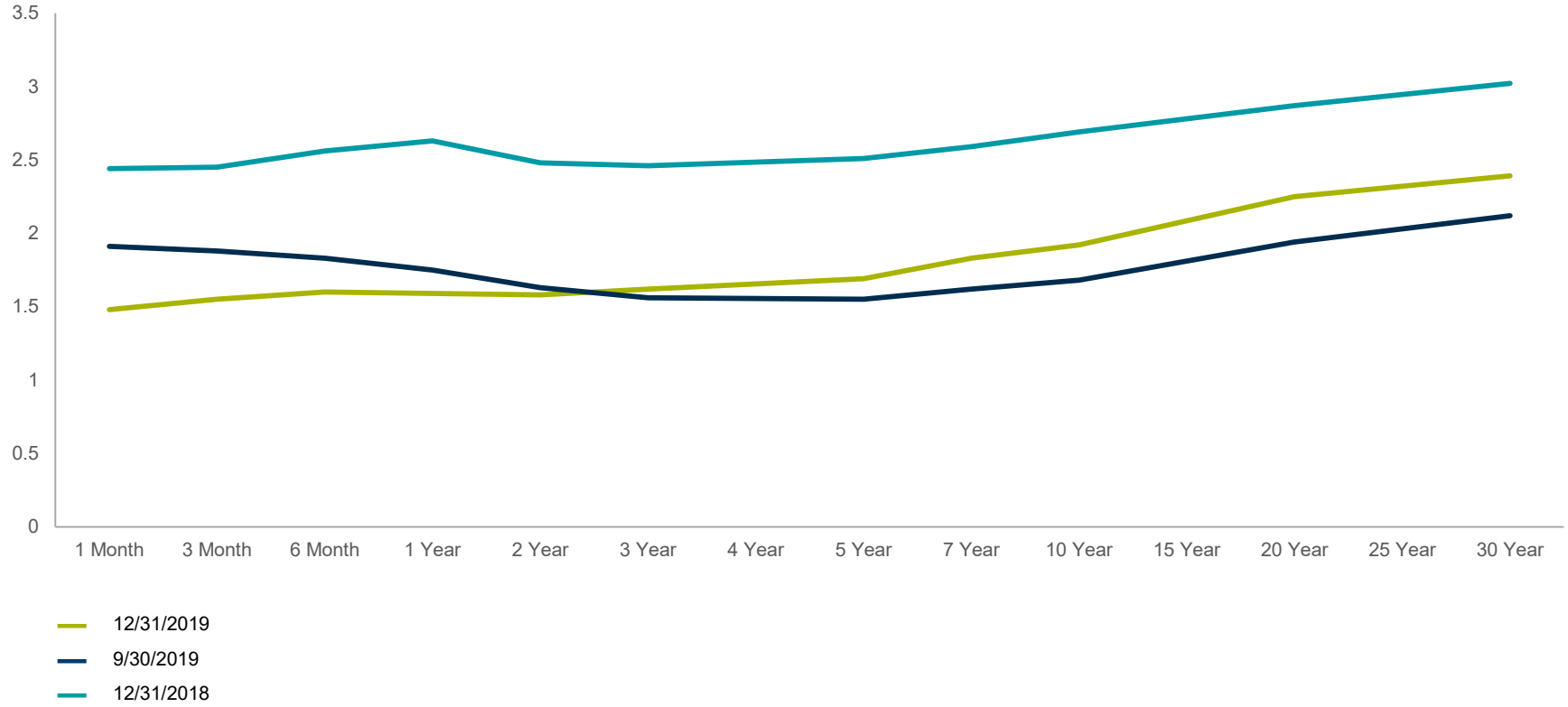


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Sources: FactSet and Vanguard.
U.S. markets measured by CRSP US Total Market Index.

Yields have fallen across the U.S. Treasury yield curve in 2019

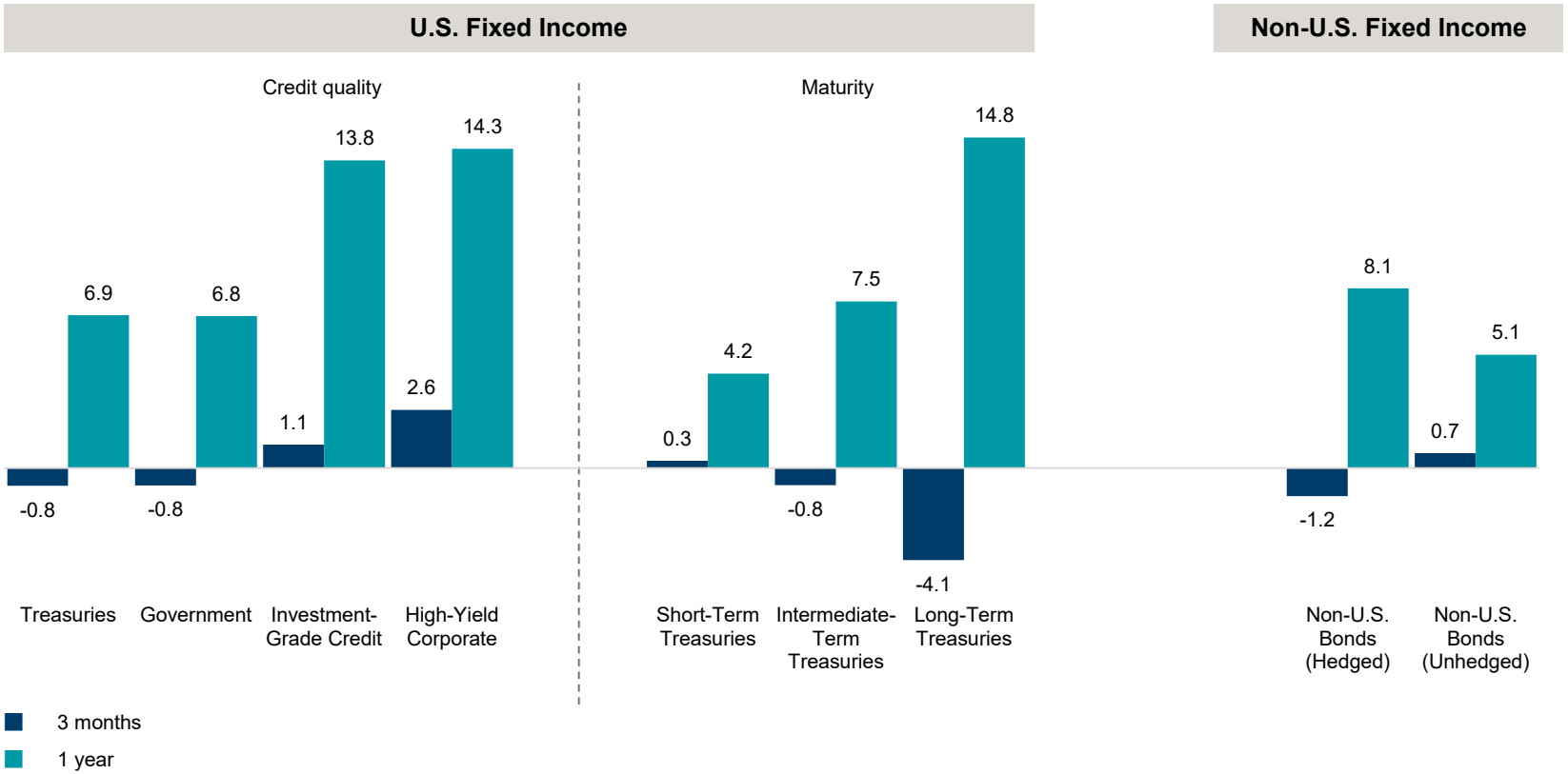
Yield (%) and change (bps)	1 month	3 month	6 month	1 year	2 year	3 year	5 year	7 year	10 year	20 year	30 year
Current yield (%)	1.48	1.55	1.60	1.59	1.58	1.62	1.69	1.83	1.92	2.25	2.39
3 month change	–43	–33	–23	–16	–5	6	14	21	24	31	27
12 month change	–96	–90	–96	–104	–90	–84	–82	–76	–77	–62	–63



Source: U.S. Treasury.

All segments of the fixed income markets had positive returns with low inflation fears in 2019

Global fixed income market returns as of December 31, 2019 (%)



Past performance is no guarantee of future returns. The performance of an index is not an exact representation of any particular investment, as you cannot invest directly in an index.

Sources: Bloomberg Barclays.
Treasuries-Government-Investment Grade Corporates-High Yield (Bloomberg Barclays US Treasury/Government/Credit/Corporate High Yield Indices); Short-Inter-Long-term Treasuries (Bloomberg Barclays US 1-5/5-10/Long Year Treasury Indices)
Non-US Bonds hedged (Bloomberg Barclays Global Aggregate ex-USD Float Adjusted RIC Capped Index hedged), Non-US Bonds unhedged (Bloomberg Barclays Global Aggregate Index ex-USD).

Portfolio performance and asset allocation

Asset allocation summary of CFMC VIAS composite portfolio
As of December 31, 2019

	Market value as of December 31, 2019	Policy target allocation	Actual weight	Variance (%)	Variance ($)
Domestic equity					
Total Stock Market Index Fund Institutional Shares	$78,602,047	39.0%	40.1%	1.1%	$2,218,840
Domestic equity subtotal	*$78,602,047*	*39.0%*	*40.1%*	*1.1%*	
International equity					
Total International Stock Index Fund Institutional Shares	$61,567,285	31.0%	31.4%	0.4%	$852,428
International equity subtotal	*$61,567,285*	*31.0%*	*31.4%*	*0.4%*	
Fixed income					
Total Bond Market Index Fund Institutional Shares	$42,008,276	22.6%	21.4%	-1.2%	-$2,254,813
Private loans*	$4,300,000	2.4%	2.2%	-0.2%	–$400,505
Fixed income subtotal	*$46,308,276*	*25.0%*	*23.6%*	*-1.4%*	
Other investments					
Real Estate Index Fund Institutional Shares	$9,376,769	5.0%	4.8%	-0.2%	-$415,950
Other investments subtotal	*$9,376,769*	*5.0%*	*4.8%*	*-0.2%*	
Total portfolio	**$195,854,377**	**100.0%**	**100.0%**		

* Private loans are not managed by Vanguard.

Community Foundation for Muskegon County
Combined fee analysis as of December 31, 2019

Investment management fee	Assets in tier	Management fee (%)	Management fee ($)
First $10 million	$10,000,000	0.15%	$15,000
Next $10 million	$10,000,000	0.12%	$12,000
Next $30 million	$30,000,000	0.08%	$24,000
Next $50 million	$50,000,000	0.03%	$15,000
Over $100 million	$91,833,371	0.02%	$18,637
Total	**$191,833,371**	**0.04%**	**$84,367**

Fund expenses	Market value	% of portfolio	Fund expense (%)	Fund expense ($)
Equity				
Total Stock Market Index Fund Institutional Shares	$78,602,047	40.97%	0.030%	$23,581
Total International Stock Index Fund Institutional Shares	$61,567,285	32.09%	0.080%	$49,254
Fixed Income				
Total Bond Market Index Fund Institutional Shares	$42,147,788	21.97%	0.035%	$14,752
Intermediate-Term Bond Index Fund Admiral™ Shares	$139,482	0.07%	0.070%	$98
Real Estate				
REIT Index Fund Institutional Shares	$9,376,769	4.89%	0.100%	$9,377
Total	**$191,833,371**	**100%**	**0.051%**	**$97,061**

Total fees and expenses	(%)	($)
Investment management fee	0.04%	$84,367
Fund expenses	0.05%	$97,061
Estimated annual fees and expenses	**0.09%**	**$181,427**

Note: Fund expenses are netted from mutual fund performance daily. Investment management fees are paid for out of pocket and are invoiced or swept quarterly.

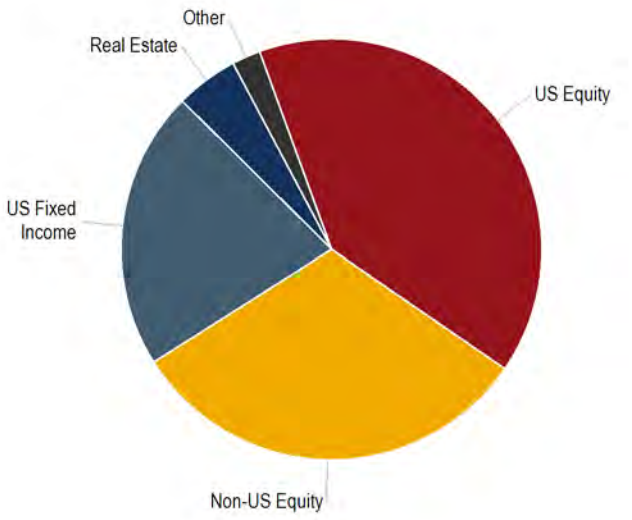
Total Portfolio Performance & Asset Allocation

Performance Summary ending December 31, 2019

	Market Value ($)	1 Mo (%)	3 Mo (%)	1 Yr (%)	3 Yrs (%)	5 Yrs (%)	10 Yrs (%)	Inception (%)	Inception Date
COMMUNITY FOUNDATION FOR MUSKEGON COUNTY	**195,854,377**	**2.48**	**6.36**	**21.72**	**9.81**	**7.27**	**--**	**7.26**	**Apr-14**
COMMUNITY FOUNDATION FOR MUSKEGON COUNTY (Net)		**2.48**	**6.35**	**21.67**	**9.76**	**7.23**	**--**	**7.21**	
Composite Benchmark		2.51	6.41	22.43	9.76	7.26	--	7.24	Apr-14

- Composite Benchmark = 39% Spliced Total Stock Market Index / 31% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 5% Real Estate Spliced Index

Current Allocation as of December 31, 2019



	Current $	Current %	Policy	Difference*
US Equity	$78,602,047	40.1%	39.0%	1.1%
Non-US Equity	$61,567,285	31.4%	31.0%	0.4%
US Fixed Income	$42,008,276	21.4%	22.6%	-1.2%
Real Estate	$9,376,769	4.8%	5.0%	-0.2%
Other**	$4,300,000	2.2%	2.4%	-0.2%
Total	**$195,854,377**	**100.0%**	**100.0%**	

*Difference between Policy and Current Allocation
** “Other” represents the foundation’s private loans, which are not being managed by Vanguard.

Gross of Advisory Fee returns reflect the deduction of fund expense ratios and any purchase or redemption fees.
Net of Fee returns reflect the deduction of fund expense ratios, any purchase or redemption fees, and VIAS advisory fee applied to the client portfolio.

Performance Summary (Gross of Advisory Fees) ending December 31, 2019

	Market Value ($)	% of Portfolio	1 Mo (%)	3 Mo (%)	1 Yr (%)	3 Yrs (%)	5 Yrs (%)	10 Yrs (%)	Inception (%)	Inception Date
COMMUNITY FOUNDATION FOR MUSKEGON COUNTY	**195,854,377**	**100.00**	**2.48**	**6.36**	**21.72**	**9.81**	**7.27**	**--**	**7.26**	**Apr-14**
COMMUNITY FOUNDATION FOR MUSKEGON COUNTY (Net)			**2.48**	**6.35**	**21.67**	**9.76**	**7.23**	**--**	**7.21**	
Composite Benchmark			2.51	6.41	22.43	9.76	7.26	--	7.24	Apr-14
MUSKEGON COUNTY SPLICED TOTAL			2.35	5.51	18.55	9.28	6.11	--	5.39	Apr-14
Total Equity	**140,169,332**	**71.57**	**3.48**	**9.03**	**26.70**	**12.49**	**8.97**	**--**	**8.54**	**Apr-14**
Equity Domestic	**78,602,047**	**40.13**	**2.87**	**9.01**	**30.81**	**14.56**	**11.21**	**--**	**11.74**	**Apr-14**
Spliced Total Stock Market Index			2.87	9.00	30.84	14.56	11.21	13.44	11.74	Apr-14
Equity International	**61,567,285**	**31.44**	**4.27**	**9.05**	**21.56**	**9.90**	**5.88**	**--**	**4.01**	**Apr-14**
Spliced Total International Stock Index			4.43	9.26	21.80	9.84	5.84	5.13	4.12	Apr-14
Total Fixed Income	**42,008,276**	**21.45**	**-0.14**	**0.03**	**8.73**	**4.03**	**3.01**	**--**	**3.19**	**Apr-14**
Fixed Income Domestic	**42,008,276**	**21.45**	**-0.14**	**0.03**	**8.73**	**4.03**	**3.01**	**--**	**3.19**	**Apr-14**
Spliced Bloomberg Barclays US Aggregate Float Adjusted Index			-0.09	0.14	8.87	4.08	3.07	3.78	3.26	Apr-14
Total Real Estate	**9,376,769**	**4.79**	**0.83**	**0.63**	**29.02**	**8.39**	**7.20**	**--**	**8.92**	**Apr-14**
Real Estate Domestic	**9,376,769**	**4.79**	**0.83**	**0.63**	**29.02**	**8.39**	**7.20**	**--**	**8.92**	**Apr-14**
Real Estate Spliced Index			0.82	0.65	29.03	8.46	7.27	12.06	9.00	Apr-14
Private Loans	**4,300,000**	**2.20**	**0.27**	**0.81**	**3.29**	**--**	**--**	**--**	**3.29**	**Mar-18**

Gross of Advisory Fee returns reflect the deduction of fund expense ratios and any purchase or redemption fees.
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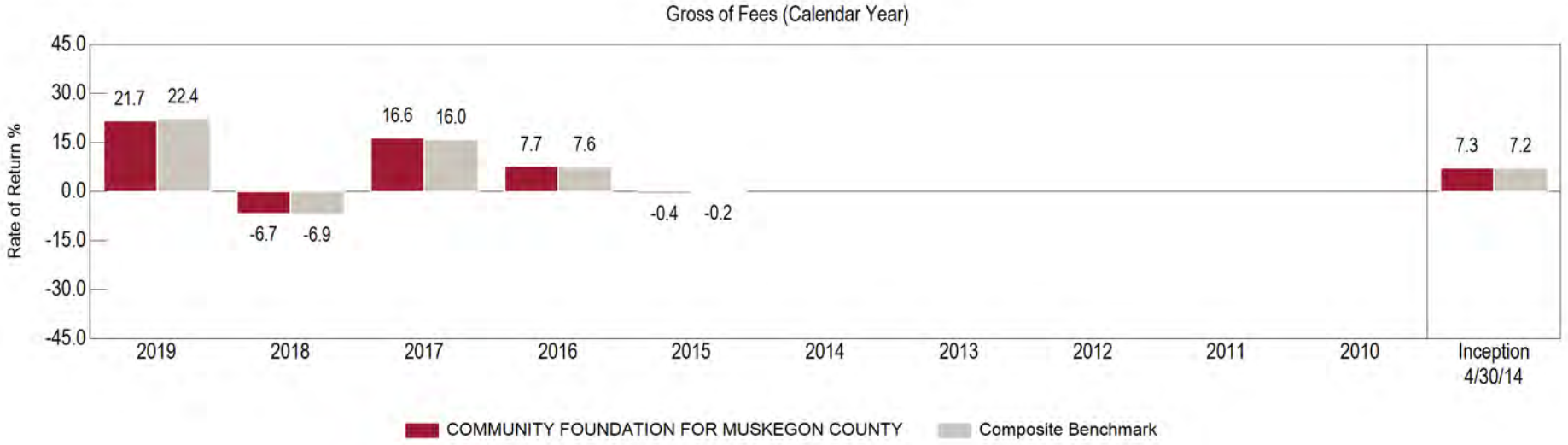
Performance Summary (Gross of Advisory Fees) ending December 31, 2019

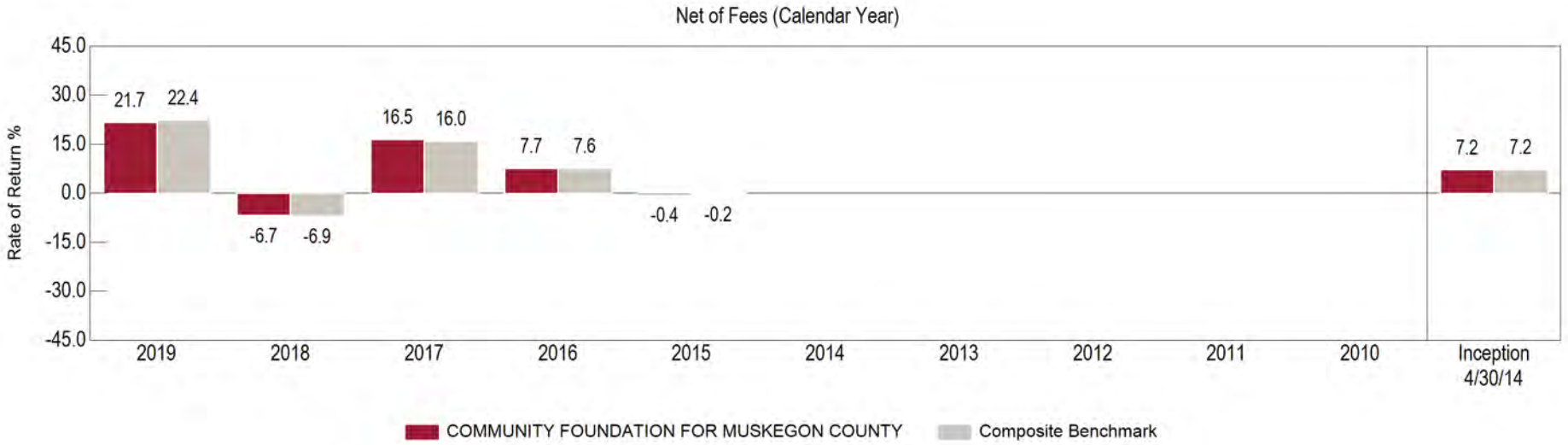
	Market Value ($)	% of Portfolio	1 Mo (%)	3 Mo (%)	1 Yr (%)	3 Yrs (%)	5 Yrs (%)	10 Yrs (%)	Inception (%)	Inception Date
COMMUNITY FOUNDATION FOR MUSKEGON COUNTY	**195,854,377**	**100.00**	**2.48**	**6.36**	**21.72**	**9.81**	**7.27**	**--**	**7.26**	**Apr-14**
Composite Benchmark			2.51	6.41	22.43	9.76	7.26	--	7.24	Apr-14
MUSKEGON COUNTY SPLICED TOTAL			2.35	5.51	18.55	9.28	6.11	--	5.39	Apr-14
Total Equity	**140,169,332**	**71.57**	**3.48**	**9.03**	**26.70**	**12.49**	**8.97**	**--**	**8.54**	**Apr-14**
Equity Domestic	**78,602,047**	**40.13**	**2.87**	**9.01**	**30.81**	**14.56**	**11.21**	**--**	**11.74**	**Apr-14**
Spliced Total Stock Market Index			2.87	9.00	30.84	14.56	11.21	13.44	11.74	Apr-14
Vanguard® Total Stock Market Index Fund Institutional Shares	78,602,047	40.13	2.87	9.01	30.81	14.55	11.21	13.43	11.73	Apr-14
Spliced Total Stock Market Index			2.87	9.00	30.84	14.56	11.21	13.44	11.74	Apr-14
Multi-Cap Core Funds Average			2.55	8.00	27.71	11.88	8.56	10.93	8.98	Apr-14
Equity International	**61,567,285**	**31.44**	**4.27**	**9.05**	**21.56**	**9.90**	**5.88**	**--**	**4.01**	**Apr-14**
Spliced Total International Stock Index			4.43	9.26	21.80	9.84	5.84	5.13	4.12	Apr-14
Vanguard® Total International Stock Index Fund Inst Shares	61,567,285	31.44	4.27	9.05	21.56	9.90	5.88	5.13	4.00	Apr-14
Spliced Total International Stock Index			4.43	9.26	21.80	9.84	5.84	5.13	4.12	Apr-14
International Funds Average			3.75	9.03	23.01	9.58	5.48	5.26	3.66	Apr-14
Total Fixed Income	**42,008,276**	**21.45**	**-0.14**	**0.03**	**8.73**	**4.03**	**3.01**	**--**	**3.19**	**Apr-14**
Fixed Income Domestic	**42,008,276**	**21.45**	**-0.14**	**0.03**	**8.73**	**4.03**	**3.01**	**--**	**3.19**	**Apr-14**
Spliced Bloomberg Barclays US Aggregate Float Adjusted Index			-0.09	0.14	8.87	4.08	3.07	3.78	3.26	Apr-14
Vanguard® Total Bond Market Index Fund Institutional Shares	42,008,276	21.45	-0.14	0.03	8.73	4.03	3.01	3.70	3.21	Apr-14
Spliced Bloomberg Barclays US Aggregate Float Adjusted Index			-0.09	0.14	8.87	4.08	3.07	3.78	3.26	Apr-14
Spliced Intermediate-Term Investment-Grade Debt Funds Average			-0.05	0.21	8.69	3.78	2.84	3.80	2.93	Apr-14
Total Real Estate	**9,376,769**	**4.79**	**0.83**	**0.63**	**29.02**	**8.39**	**7.20**	**--**	**8.92**	**Apr-14**
Real Estate Domestic	**9,376,769**	**4.79**	**0.83**	**0.63**	**29.02**	**8.39**	**7.20**	**--**	**8.92**	**Apr-14**
Real Estate Spliced Index			0.82	0.65	29.03	8.46	7.27	12.06	9.00	Apr-14
Vanguard® Real Estate Index Fund Institutional Shares	9,376,769	4.79	0.83	0.63	29.02	8.39	7.20	12.01	8.92	Apr-14
Real Estate Spliced Index			0.82	0.65	29.03	8.46	7.27	12.06	9.00	Apr-14
Real Estate Funds Average			-0.22	0.43	26.64	7.90	6.52	11.12	8.17	Apr-14

Performance Summary (Gross of Advisory Fees) ending December 31, 2019

	Market Value ($)	% of Portfolio	1 Mo (%)	3 Mo (%)	1 Yr (%)	3 Yrs (%)	5 Yrs (%)	10 Yrs (%)	Inception (%)	Inception Date
Private Loans	**4,300,000**	**2.20**	**0.27**	**0.81**	**3.29**	**--**	**--**	**--**	**3.29**	**Mar-18**
Private Loan 1	2,300,000	1.17	0.27	0.81	3.29	--	--	--	3.29	Mar-18
Private Loan 2	2,000,000	1.02	0.27	0.81	3.29	--	--	--	3.29	Mar-18

Total Portfolio Performance





Gross of Advisory Fee returns reflect the deduction of fund expense ratios and any purchase or redemption fees.
Net of Fee returns reflect the deduction of fund expense ratios, any purchase or redemption fees, and VIAS advisory fee applied to the client portfolio.

Cash Flow Summary

	Last Month	Last Three Months	Year-To-Date	One Year
Beginning Market Value	$191,805,832	$186,963,410	$164,471,143	$164,471,143
Net Cash Flow	-$691,285	-$2,853,128	-$4,103,418	-$4,103,418
Capital Appreciation	$3,427,727	$10,223,307	$30,607,677	$30,607,677
Income	$1,312,103	$1,520,788	$4,878,975	$4,878,975
Ending Market Value	$195,854,377	$195,854,377	$195,854,377	$195,854,377

Month Ending December 31, 2019

	Beginning Market Value	Net Cash Flow	Capital Appreciation	Income	Ending Market Value
Private Loan 1	$2,300,000	-$6,229	$0	$6,229	$2,300,000
Private Loan 2	$2,000,000	-$5,417	$0	$5,417	$2,000,000
Vanguard® Real Estate Index Fund Institutional Shares	$9,299,516	$0	-$18,793	$96,046	$9,376,769
Vanguard® Total Bond Market Index Fund Institutional Shares	$42,065,688	$0	-$151,725	$94,313	$42,008,276
Vanguard® Total International Stock Index Fund Inst Shares	$59,046,494	$0	$1,833,563	$687,228	$61,567,285
Vanguard® Total Stock Market Index Fund Institutional Shares	$77,094,135	-$679,639	$1,764,683	$422,870	$78,602,047
Total	**$191,805,832**	**-$691,285**	**$3,427,727**	**$1,312,103**	**$195,854,377**

Investment Expense Analysis as of December 31, 2019

Name	Market Value	% of Portfolio	Expense Ratio
Total Equity	**$140,169,332**	**71.6%**	
Equity Domestic	**$78,602,047**	**40.1%**	
Vanguard® Total Stock Market Index Fund Institutional Shares	$78,602,047	40.1%	0.030%
Equity International	**$61,567,285**	**31.4%**	
Vanguard® Total International Stock Index Fund Inst Shares	$61,567,285	31.4%	0.080%
Total Fixed Income	**$42,008,276**	**21.4%**	
Fixed Income Domestic	**$42,008,276**	**21.4%**	
Vanguard® Total Bond Market Index Fund Institutional Shares	$42,008,276	21.4%	0.035%
Total Real Estate	**$9,376,769**	**4.8%**	
Real Estate Domestic	**$9,376,769**	**4.8%**	
Vanguard® Real Estate Index Fund Institutional Shares	$9,376,769	4.8%	0.100%
Private Loans	**$4,300,000**	**2.2%**	
Private Loan 1	$2,300,000	1.2%	
Private Loan 2	$2,000,000	1.0%	
Total	**$195,854,377**	**100.0%**	**0.049%**

Market Performance as of December 31, 2019

Name	Dec-19	Last 3 Months	1 Yr	3 Yrs	5 Yrs	10 Yrs
US Equity						
CRSP US Total Market TR USD	2.9	9.0	30.8	14.6	11.2	13.4
S&P 500	3.0	9.1	31.5	15.3	11.7	13.6
S&P 400 MidCap	2.8	7.1	26.2	9.3	9.0	12.7
S&P 600 SmallCap	3.0	8.2	22.8	8.4	9.6	13.4
International Equity						
MSCI Emerging Markets	7.5	11.8	18.4	11.6	5.6	3.7
MSCI Emerging Markets NR LCL	5.7	9.5	18.1	11.5	7.5	6.1
MSCI EAFE	3.2	8.2	22.0	9.6	5.7	5.5
MSCI EAFE NR LCL	1.3	5.2	21.7	7.7	6.7	7.2
MSCI ACWI ex USA	4.3	8.9	21.5	9.9	5.5	5.0
Fixed Income Domestic						
BBgBarc US Aggregate TR	-0.1	0.2	8.7	4.0	3.0	3.7
BBgBarc US Corporate 1-5 Years TR	0.4	1.0	7.0	3.5	2.9	3.3
BBgBarc US Credit/Corp 5-10 Yr TR	0.5	1.3	14.3	5.9	4.8	5.9
BBgBarc US Corporate Long TR	0.1	1.3	23.9	8.8	6.4	8.0
BBgBarc US Govt/Credit Long TR	-1.1	-1.1	19.6	8.1	5.4	7.6
BBgBarc US Treasury Strips 20-30 Yr Equal Parity TR	-4.0	-5.9	20.1	9.3	4.9	10.1
BBgBarc US High Yield TR	2.0	2.6	14.3	6.4	6.1	7.6
BBgBarc US Govt TR	-0.6	-0.8	6.8	3.3	2.4	3.0
BBgBarc US Credit TR	0.3	1.1	13.8	5.8	4.4	5.3
BBgBarc US Treasury 1-5 Yr TR	0.1	0.3	4.2	2.1	1.7	1.7
BBgBarc US Treasury 5-10 Yr TR	-0.4	-0.8	7.5	3.6	2.7	4.0
BBgBarc US Treasury Long TR	-2.8	-4.1	14.8	6.9	4.1	7.0
BBgBarc US Treasury TIPS 0-5 Yr TR	0.7	1.1	4.9	2.1	1.8	1.6
BBgBarc US TIPS TR	0.4	0.8	8.4	3.3	2.6	3.4
Fixed Income International						
BBgBarc Global Aggregate ex US Tres Hedged TR	-0.3	-1.1	7.6	4.4	3.9	4.3
BBgBarc Emerging Markets TR	1.5	2.1	13.1	6.1	5.8	6.6
REIT						
MSCI US REIT Gross	-0.6	-0.8	25.8	8.1	7.0	11.9

Benchmark History as of December 31, 2019

COMMUNITY FOUNDATION FOR MUSKEGON COUNTY		
8/1/2018	Present	39% Spliced Total Stock Market Index / 31% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 5% Real Estate Spliced Index
7/1/2018	7/31/2018	39% Spliced Total Stock Market Index / 30% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 6% Real Estate Spliced Index
6/1/2018	6/30/2018	39% Spliced Total Stock Market Index / 29% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 7% Real Estate Spliced Index
4/1/2018	5/31/2018	39% Spliced Total Stock Market Index / 28% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 8% Real Estate Spliced Index
3/1/2018	3/31/2018	39% Spliced Total Stock Market Index / 27% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 9% Real Estate Spliced Index
9/1/2017	2/28/2018	39% Spliced Total Stock Market Index / 26% Spliced Total International Stock Index / 25% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 10% REIT Spliced Index
8/1/2017	8/31/2017	38% Spliced Total Stock Market Index / 25.33% Spliced Total International Stock Index / 26.67% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 10% REIT Spliced Index
7/1/2017	7/31/2017	37.5% Spliced Total Stock Market Index / 25% Spliced Total International Stock Index / 27.5% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 10% REIT Spliced Index
5/1/2014	6/30/2017	10% REIT Spliced Index / 30% Spliced Bloomberg Barclays US Aggregate Float Adjusted Index / 24% Spliced Total International Stock Index / 36% Spliced Total Stock Market Index
Total Equity		
		N/A
Equity Domestic		
4/30/2014	Present	100% CRSP US Total Market TR USD
Vanguard® Total Stock Market Index Fund Institutional Shares		
4/30/2014	Present	100% CRSP US Total Market TR USD
Equity International		
4/30/2014	Present	100% FTSE Global All-Cap ex-US Index
Vanguard® Total International Stock Index Fund Inst Shares		
4/30/2014	Present	100% FTSE Global All-Cap ex-US Index

Benchmark History as of December 31, 2019

Total Fixed Income		
		N/A
Fixed Income Domestic		
4/30/2014	Present	100% BBgBarc US Aggregate Float Adjusted TR
Vanguard® Total Bond Market Index Fund Institutional Shares		
4/30/2014	Present	100% BBgBarc US Aggregate Float Adjusted TR
Total Real Estate		
		N/A
Real Estate Domestic		
7/25/2018	Present	100% MSCI US IM Real Estate 25/50 Index
1/26/2018	7/24/2018	100% MSCI US IM Real Estate 25/50 Tran Index
4/30/2014	1/25/2018	100% MSCI US REIT Gross
Vanguard® Real Estate Index Fund Institutional Shares		
7/25/2018	Present	100% MSCI US IM Real Estate 25/50 Index
1/26/2018	7/24/2018	100% MSCI US IM Real Estate 25/50 Tran Index
4/30/2014	1/25/2018	100% MSCI US REIT Gross
Private Loans		
		N/A
Private Loan 1		
		N/A
Private Loan 2		
		N/A

Quarterly summary of combined historical performance
As of December 31, 2019

	Annualized return					Calendar return								
	QTR	1 year	3 year	5 year	10 year	2018	2017	2016	2015	2014	2013	2012	2011	2010
CFMC Composite Portfolio	6.35	21.67	9.76	7.23	7.58	–6.71	16.50	7.67	–0.44	5.68	12.35	12.21	–3.47	13.95
*CFMC Composite Benchmark**	6.41	22.44	9.76	7.26	8.82	–6.87	15.96	7.58	–0.22	7.08	19.26	12.92	0.70	12.98

Past performance is no guarantee of future returns. The performance of an index is not an exact representation of any particular investment, as you cannot invest directly in an index.

Notes:
Composite performance data consists of previous providers' portfolio and benchmark returns through April 2014; VIAS returns thereafter. Data prior to May 2014 is based upon information taken from the previous providers' quarterly performance reports and is provided as an estimate only. Composite does not include segregated accounts or pledge holdings.
Return calculations are derived from Morningstar Direct.
* Composite Benchmark allocations over time:
50% S&P 500 / 10% Russell 2000 / 10% MSCI EAFE / 30% LB AGG from March 2004 to March 2005;
45% S&P 500 / 12.5% Russell 2000 / 12.5% MSCI EAFE / 30% LB AGG from March 2005 to December 2007;
55% Russell 3000 / 15% MSCI AC World Index ex-US / 30% LB AGG from December 2007 to December 2008;
55% Russell 3000 / 15% MSCI AC World Index ex-US / 30% Barclays AGG from December 2008 to April 2014;
36% Spliced Total Stock Market Index (CRSP US Total Market Index) / 30% Spliced Barclays US Agg Float-Adj Ix (Barclays U.S. Aggregate Float Adjusted Index)/24% Spliced Total Int'l Stock Index (FTSE Global All Cap ex US Index) / 10% REIT Spliced Index (MSCI US REIT Index) from May 2014 to June 2017.
39% Spliced Total Stock Market Index (CRSP US Total Market Index) / 25% Spliced Barclays US Agg Float-Adj Ix (Barclays U.S. Aggregate Float Adjusted Index)/26% Spliced Total Int'l Stock Index (FTSE Global All Cap ex US Index) / 10% REIT Spliced Index (MSCI US REIT Index) from July 2017 to February 2018.
39% Spliced Total Stock Market Index (CRSP US Total Market Index) / 25% Spliced Barclays US Agg Float-Adj Ix (Barclays U.S. Aggregate Float Adjusted Index)/31% Spliced Total Int'l Stock Index (FTSE Global All Cap ex US Index) / 5% REIT Spliced Index (MSCI US REIT Index) from March 2018 to the present.

Legal

For more information about Vanguard funds or non-Vanguard funds offered through Vanguard Brokerage Services, visit vanguard.com or call your Investment Consultant or Relationship Manager to obtain a prospectus or, if available, a summary prospectus. Investment objectives, risks, charges, expenses, and other important information are contained in the prospectus; read and consider it carefully before investing. Total Portfolio Net of Fees returns reflect the deduction of fund expense ratios, purchase or redemption fees, and any advisory service fee applied to the client portfolio.

Total Portfolio returns represent client-specific time-weighted returns (TWR) are presented gross of any applicable service fees with the exception of mutual fund expense ratios and other security-level expenses.

Internal rates of return (IRR) are net of any applicable service fees, include account-specific cash flows, and are not directly comparable to a benchmark, since benchmarks do not include cash flows.

Client performance inception date is generally the first month-end after initial funding. Mutual funds and all investments are subject to risk, including the possible loss of the money you invest. Diversification does not ensure a profit or protect against a loss.

Performance figures assume the reinvestment of dividends and capital gains distributions. The fund performance percentages are based on fund total return data, adjusted for expenses, obtained from Lipper, a Thomson Reuters Company. The total return data was not adjusted for fees and loads.

Benchmark comparative indexes represent unmanaged or average returns on various financial assets, which can be compared with funds' total returns for the purpose of measuring relative performance.

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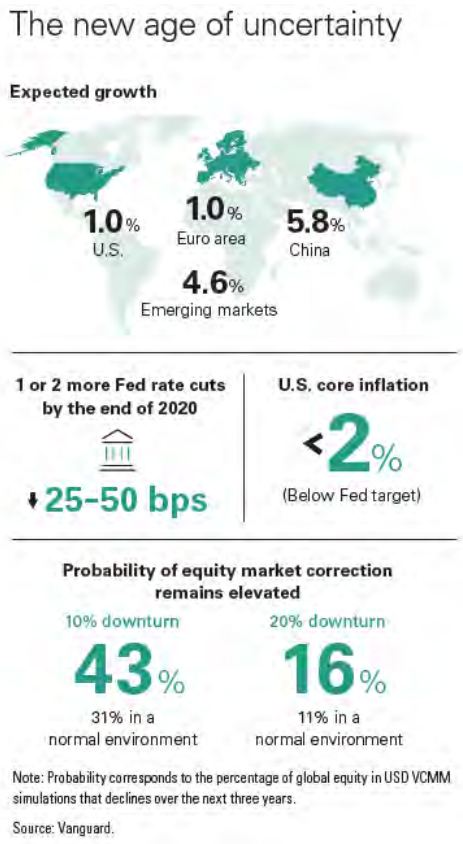
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Economic and market outlook

Vanguard 2020 market and economic outlook at a glance



Key take aways:

- Increasingly unpredictable policy environment is undermining economic activity.
- Global growth to slow further but we're a long way from a serious global contraction.
- The chance of a large drawdown in stocks and other high-beta assets remains elevated.

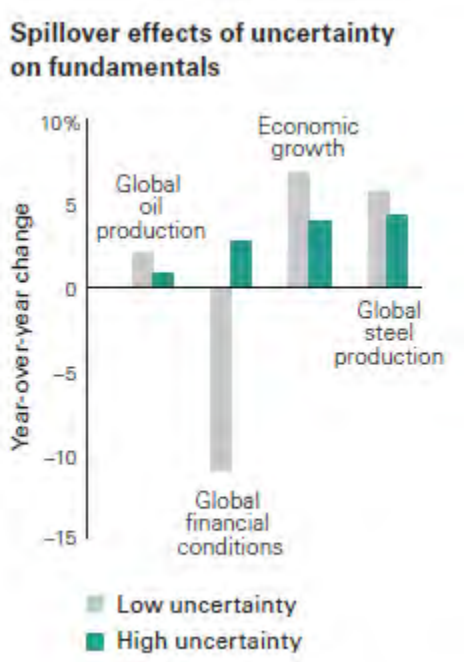
Investment outlook

The 10-year annualized nominal return projections, based on market conditions as of September 30, 2019, are as follows:

Equities	
U.S. equities	3.5%–5.5%
U.S. growth	2.5%–4.5%
U.S. value	6.0%–8.0%
U.S. large-cap	3.5%–5.5%
U.S. small-cap	4.5%–6.5%
U.S. real estate investment trusts	2.5%–4.5%
Global equities ex-U.S. (unhedged)	6.5%–8.5%
Fixed income	
U.S. bonds	2.0%–3.0%
U.S. Treasury bonds	1.5%–2.5%
U.S. credit bonds	2.5%–3.5%
U.S. high-yield corporate bonds	3.0%–4.0%
U.S. Treasury inflation-protected securities	1.0%–2.0%
U.S. cash	1.5%–2.5%
Global bonds ex-U.S. (hedged)	1.5%–2.5%

These probabilistic return assumptions depend on current market conditions and, as such, may change over time.
IMPORTANT: The projections and other information generated by the Vanguard Capital Markets Model® regarding the likelihood of various investment outcomes are hypothetical in nature, do not reflect actual investment results, and are not guarantees of future results. Distribution of return outcomes from the VCMM are derived from 10,000 simulations for each modeled asset class. Simulations are as of September 30, 2019.
Results from the model may vary with each use and over time. For more information, see page 4.
Source: Vanguard Investment Strategy Group.

Vanguard 2020 market and economic outlook at a glance



Trade tensions and unpredictable policymaking take their toll

An increasingly unpredictable policy environment is undermining economic activity globally through postponed investments and declines in production. But we don't seem on the verge of a serious global contraction.

We do not foresee a significant reversal of trade tensions or expect that policymaking will become more predictable in the year ahead. For example, Vanguard estimates there's only a 10% chance of an upside scenario where China and the U.S. sign a series of trade deals, roll back tariffs, and continue negotiations on structural issues.

Our research also shows that periods of high uncertainty are associated with lower growth, tighter financial conditions, and lower asset returns.

4%

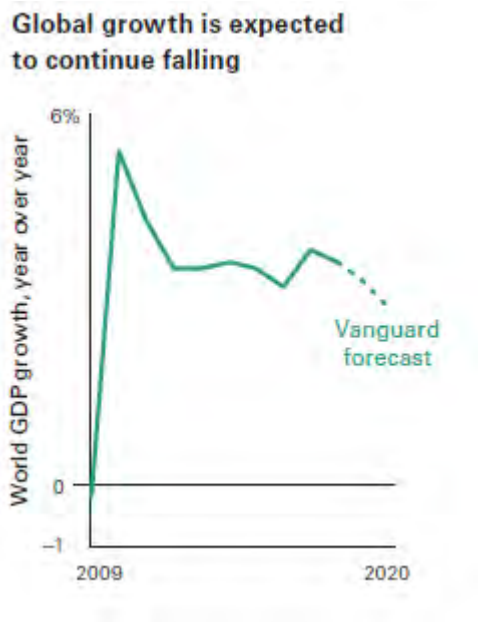
We estimate that in periods of **high uncertainty,** year-over-year global growth averages around 4%.

7%

We estimate that in periods of **low uncertainty,** year-over-year global growth averages around 7%.

Notes: The bars represent the average year-over-year change in each of the indicators in high- versus low-uncertainty periods. Periods of low versus high uncertainty are obtained through a Markov-switching model for global growth. Global financial conditions are an aggregate measure of risk sentiment and include variables such as equity returns, credit spreads, and lending behavior. Lower values denote easier financial conditions and risk-on attitudes. Z-scores measure how far a value differs from the historical average, accounting for the measure's typical fluctuations.
Sources: Vanguard calculations, based on data from Moody's Analytics Data Buffet and Thomson Reuters Datastream.

Vanguard 2020 market and economic outlook at a glance



Growth: Prospects around the globe are deteriorating

United States: Downshifting for an uncertain road ahead
Vanguard expects elevated levels of uncertainty to persist through at least 2020 and continue to weigh on business sentiment, along with waning support from the consumer. We see U.S. growth falling below trend to around 1% in 2020, with the risk of recession still elevated.

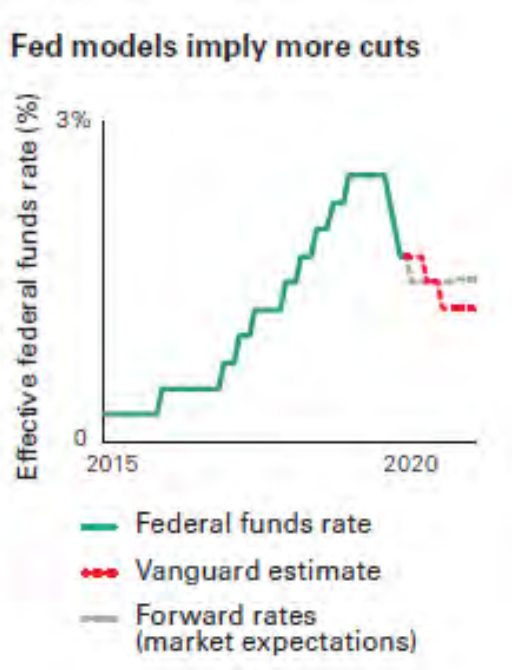
China: No hard landing, uncertainty impedes stimulus
Slowing global economic growth and expectations for U.S.-China tensions place China's economy in an environment of perpetual high policy uncertainty. These factors constitute a sizable headwind for both immediate and medium-term growth prospects. China will likely slow to a below-trend pace of 5.8% in 2020.

Euro area: No strong rebound in sight given limited fiscal stimulus
The euro-area economy has continued to slow because of the importance of industrial trade to its economy and some drag from Brexit-related uncertainty. Growth in the euro area is likely to stay weak at around 1%.

Emerging markets: Global trade slowdown
Economic growth for emerging markets in the aggregate is expected to be 4.6% in 2020. However, we expect there to be vast heterogeneity both within and among regions. Some of the recent slowdown across emerging markets reflects the spillover effects of a slowing China, policy tightening by the U.S. Federal Reserve in 2018, and a decline in global trade.

Sources: International Monetary Fund and Vanguard forecasts.

Vanguard 2020 market and economic outlook at a glance



Monetary policy: The pivot to looser policy continues

Despite increased doubts about the effectiveness of monetary policy, we expect central banks to continue to adopt unconventional measures, while significant fiscal stimulus remains unlikely unless there is a more severe downturn. Our outlook is for subdued inflation trends across major economies, consistent with the inflation expectations held by consumers and financial markets.

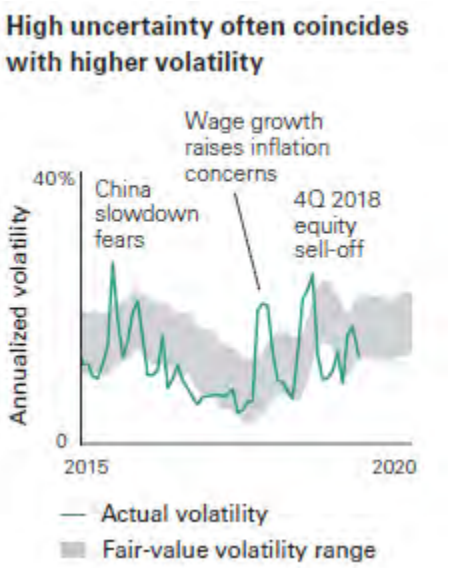
United States. Vanguard expects that the U.S. economy, and in turn the U.S. Federal Reserve, will shift into a lower gear in 2020 as policy-makers, businesses, and consumers navigate a more uncertain road ahead. That translates into the Fed reducing the federal funds rate by 25 to 50 basis points before the end of 2020. We expect inflation to barely reach 2% in the U.S., with the Federal Reserve's core inflation gauge staying below its 2% policy target.

Euro area. In 2020 we expect the European Central Bank to leave policy broadly unchanged, with risks skewed toward further easing.

China. Policy efforts to stabilize growth will continue, but concerns about medium-term financial stability risks will keep these measures in moderation relative to prior easing cycles, reducing the tailwinds for global growth prospects.

Notes: Vanguard's base case expectations for 2020 (0.5%–1.5% GDP growth, 1.8% core PCE, and 100,000 new non-farm payroll jobs per month for the duration of 2020). Vanguard estimate includes model-based expectations and those of Vanguard's Investment Strategy Group.
Sources: Vanguard calculations, based on data from the U.S. Bureau of Economic Analysis, the Federal Open Market Committee, Bloomberg, and Moody's Analytics Data Buffet.

Vanguard 2020 market and economic outlook at a glance



Investment outlook: Subdued returns ahead

In the face of elevated uncertainty and a synchronized global slowdown, equity markets have remained surprisingly robust. However, investors should caution themselves against extrapolating present gains into the future.
As global growth slows further in 2020 and policy uncertainties continue, investors should expect periodic bouts of volatility and a fragile backdrop for markets in 2020 and beyond.

Near-term. Our near-term outlook for global equity markets remains guarded, and the chance of a large drawdown for equities and other high-beta assets remains elevated and significantly higher than it would be in a normal market environment. High-quality fixed income assets, whose expected returns are positive only in nominal terms, remain a key diversifier in a portfolio.

Long-term. Returns over the next decade are anticipated to be modest at best. Our expectation for fixed income returns has fallen because of declining policy rates, lower yields across maturities, and compressed corporate spreads.

10-year annualized nominal return projections

2.0%–3.0%	1.5%–2.5%	3.5%–5.5%	6.5%–8.5%
U.S. bonds	Global bonds ex-U.S. (hedged)	U.S. equities	Global equities ex-U.S. (unhedged)

Based on simulated ranges of portfolio returns and volatility, the diversification benefits of global fixed income and global equity remain compelling.

Notes: Fair-value volatility range is calculated with an OLS regression using Vanguard's leading economic indicator index (VLEI), financial conditions index (VFCI), and policy uncertainty index as independent variables. Volatility is measured as the standard deviation of daily returns of the S&P 500 index on a 30-day rolling time period, annualized. The forecasted range of volatility for 2020 is based on Vanguard's economic projections.
Sources: Vanguard calculations, based on data from Thomson Reuters Datastream and policyuncertainty.com.

Appendix
– Segregated accounts reporting

Quarterly investment performance summary of segregated accounts
For the period ending December 31, 2019

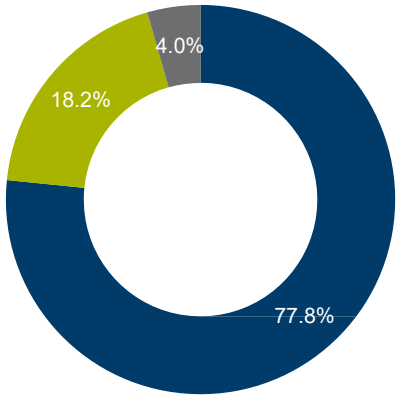
Performance net of investment manager fees

Account name	Investment manager	Q4 2019 return	2019	Market value as of 12/31/2019
Freedom - Willard Bosma	RJ	+5.88%	+19.02%	$493,435
J Hanna/M Murphy	UBS	+5.93%	+20.56%	$1,347,665
Ernest E Settle	UBS	+5.77%	+20.06%	$1,227,077
Cutler	Schwab	+5.92%	+14.86%	$206,677
The 2012 Fund	Schwab	+0.02%	+0.24%	$582,180
MI Heritage Trails (MICHHT)	UBS	+5.68%	+19.38%	$1,380,619
Donahue	NW	+0.87%	+8.32%	$231,995
G & B Hilt Fund	UBS	+5.67%	+19.33%	$7,315,488
Collins	RJ	+5.62%	+22.79%	$779,028
Campbell Scholarship	ML	+4.65%	+15.00%	$36,769
Shelby Rotary	LPL	+4.91%	+18.02%	$373,423

Note: Vanguard cannot independently validate the accuracy of the returns shown above. All returns are calculated using data as reported by CFMC staff. Return calculation reflects beginning period and ending period market values adjusted for investment provider or CFMC cash flows, and investment provider fees.

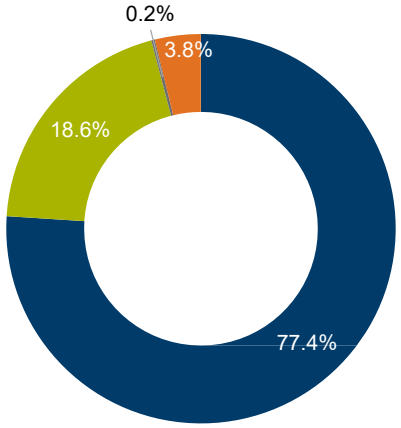
Quarterly asset allocation summary of segregated accounts
As of December 31, 2019

RJ – Freedom – Willard Bosma ($493,435)



- Equity
- Fixed Income
- Cash / Cash Alternatives

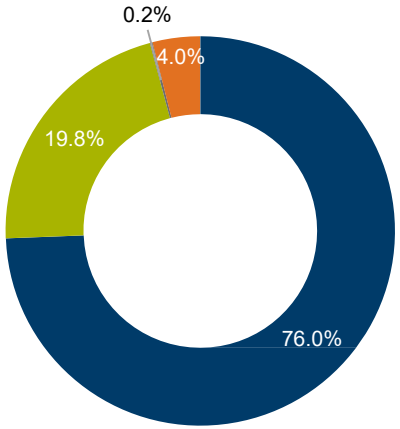
UBS – J Hanna/M Murphy ($1,347,665)



- Equity
- Fixed Income
- Cash / Cash Alternatives
- Non-classified

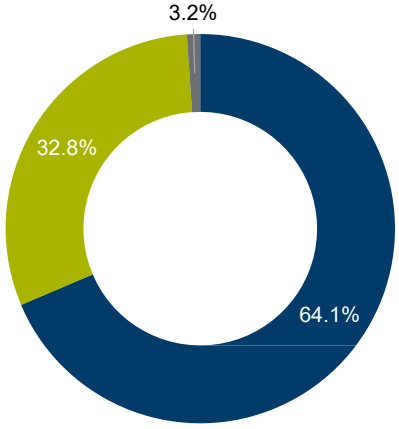
Quarterly asset allocation summary of segregated accounts
As of December 31, 2019

UBS – Ernest E Settle ($1,227,380)



- Equity
- Fixed Income
- Cash / Cash Alternatives
- Non-classified

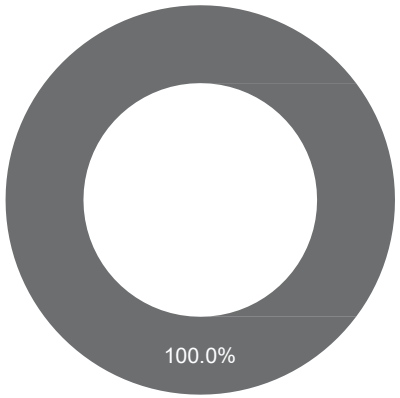
Schwab – Cutler ($195,813)



- Equity
- Fixed Income
- Cash / Cash Alternatives

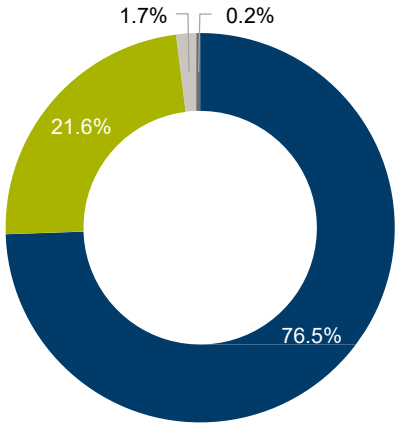
Quarterly asset allocation summary of segregated accounts
As of December 31, 2019

Schwab – The 2012 Fund ($582,180)



- Cash / Cash Alternatives

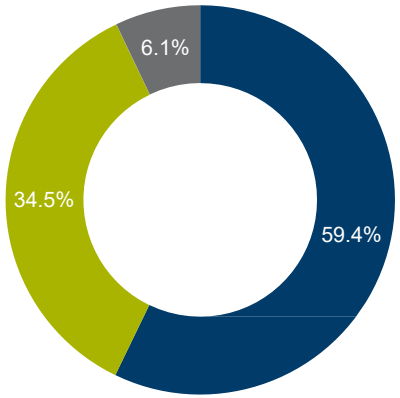
UBS - MI Heritage Trails (MICHHT) ($1,380,619)



- Equity
- Fixed Income
- Alternative Investments
- Cash / Cash Alternatives

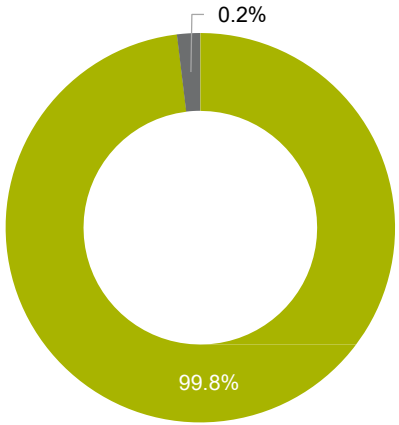
Quarterly asset allocation summary of segregated accounts
As of December 31, 2019

ML – Campbell Scholarship ($36,769)



- Equity
- Fixed Income
- Cash / Cash Alternatives

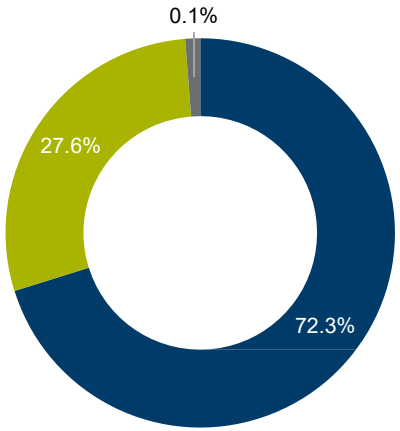
NW – Donahue ($231,995)



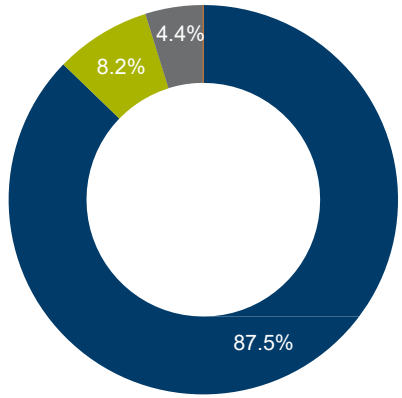
- Fixed Income
- Cash / Cash Alternatives

Quarterly asset allocation summary of segregated accounts
As of December 31, 2019

UBS – G & B Hilt Fund (hilt05) ($7,315,488)

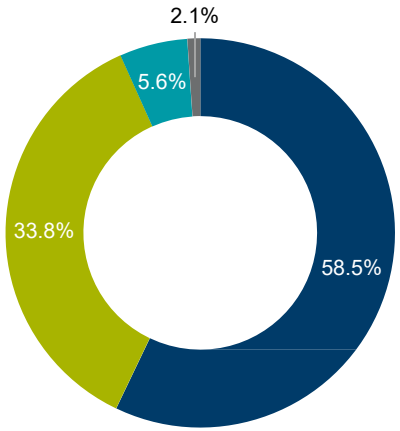


RJ – Collins ($779,028)



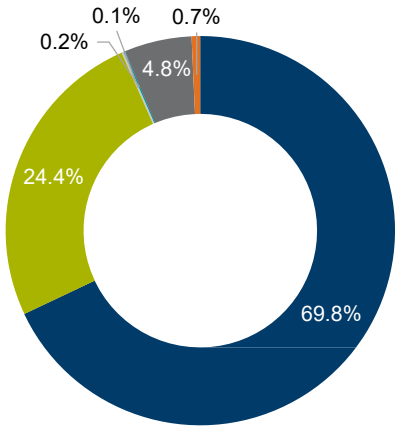
Quarterly asset allocation summary of segregated accounts
As of December 31, 2019

LPL – Shelby Rotary ($373,423)



- Equity
- Fixed Income
- Real Estate
- Cash / Cash Alternatives

Combined segregated accounts ($13,974,659)



- Equity
- Fixed Income
- Alternative Investments
- Real Estate
- Cash / Cash Alternatives
- Non-classified

Spliced benchmark information

Spliced Total Stock Market Index: Dow Jones Wilshire 5000 Index through April 22, 2005; MSCI US Broad Market Index through June 2, 2013; CRSP US Total Market Index thereafter.

Spliced Total International Stock Index: Total International Composite Index through August 31, 2006; MSCI EAFE + Emerging Markets Index through December 15, 2010; MSCI ACWI ex USA IMI Index through June 2, 2013, FTSE Global All Cap ex US Index thereafter.

REIT Spliced Index: MSCI US REIT Index adjusted to include a 2% cash position (Lipper Money Market Average) through April 30, 2009; 100% MSCI US REIT Index thereafter.

Spliced Bloomberg Barclays U.S. Aggregate Float Adjusted Index: BloomBarc U.S. Aggregate Bond Index through December 31, 2009; BloomBarc U.S. Aggregate Float Adjusted Index thereafter.

Appendix
– Fund pages

Vanguard Total Stock Market Index Fund (VITSX)

Equity characteristics

	VITSX	Benchmark
Number of stocks	3,579	3,513
Median market cap	$83.0B	$83.0B
P/E ratio (trailing earnings)	22.6x	22.5x
P/B ratio	3.2x	3.2x
Return on equity (5-year average)	15.7%	15.7%
Earnings growth rate (5-year)	10.8%	10.8%
Equity yield (dividend)	1.7%	1.7%
Foreign holdings	0.4%	—
Turnover (fiscal year end)	3.5%	—
Short-term reserves	0.0%	N/A

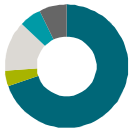
Sector diversification as a % of common stock



Sector	VITSX	Benchmark
Technology	21.8%	21.8%
Financials	19.4	19.5
Consumer Services	13.3	13.3
Industrials	13.2	13.1
Health Care	13.0	13.1
Consumer Goods	7.9	7.9
Oil & Gas	4.1	4.0
Utilities	3.2	3.2
Basic Materials	2.2	2.2
Telecommunications	1.9	1.9

Sector categories are based on the Industry Classification Benchmark ("ICB"), except for the "Other" category (if applicable), which includes securities that have not been provided an ICB classification as of the effective reporting period.

Market-cap breakdown



Market capitalization	VITSX	Benchmark
More than $31	69.7%	69.6%
$25 to $31	4.0	4.0
$9 to $25	13.4	13.4
$5 to $9	5.6	5.6
Below $5	7.3	7.3

Risk and volatility

	R-squared	Beta	Alpha	Standard deviation	Sharpe ratio
VITSX	N/A	N/A	-0.00	12.38	1.04
Primary benchmark	1.00	1.00	N/A	12.38	1.04
Broad-based benchmark	1.00	1.00	N/A	12.39	1.04

R-squared and beta are calculated from trailing 36-month fund returns relative to the associated benchmark.
Broad-based benchmark: Dow Jones U.S. Total Stock Market Float Adjusted Index.

Ten largest holdings

	% of total net assets
Microsoft Corp.	3.8%
Apple Inc.	3.7
Alphabet Inc.	2.5
Amazon.com Inc.	2.4
Facebook Inc.	1.6
Berkshire Hathaway Inc.	1.4
JPMorgan Chase & Co.	1.3
Johnson & Johnson	1.2
Visa Inc.	1.0
Procter & Gamble Co.	1.0
Top ten as a % of total net assets	**19.9%**

The holdings listed exclude any temporary cash investments and equity index products.

All data as of December 31, 2019, unless otherwise noted.

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Vanguard Total International Stock Index Fund (VTSNX)

Equity characteristics

	VTSNX	Benchmark
Number of stocks	7,423	7,069
Median market cap	$28.6B	$28.5B
P/E ratio (trailing earnings)	15.4x	15.5x
P/B ratio	1.6x	1.7x
Return on equity (5-year average)	12.4%	12.4%
Earnings growth rate (5-year)	9.7%	9.7%
Equity yield (dividend)	3.0%	3.0%
Turnover (fiscal year end)	3.5%	—
Short-term reserves	0.0%	N/A

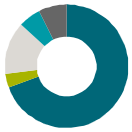
Sector diversification as a % of common stock



Sector	VTSNX	Benchmark
Financials	24.4%	24.4%
Industrials	14.7	14.7
Consumer Goods	13.9	14.0
Consumer Services	9.4	9.4
Technology	9.2	9.1
Health Care	8.7	8.7
Basic Materials	6.8	6.8
Oil & Gas	6.3	6.3
Telecommunications	3.3	3.3
Utilities	3.3	3.3

Sector categories are based on the Industry Classification Benchmark ("ICB"), except for the "Other" category (if applicable), which includes securities that have not been provided an ICB classification as of the effective reporting period.

Market-cap breakdown



Market capitalization	VTSNX	Benchmark
More than $12	69.4%	69.2%
$10 to $12	3.6	3.6
$4 to $10	13.8	13.8
$2 to $4	5.9	5.9
Below $2	7.4	7.5

Risk and volatility

	R-squared	Beta	Alpha	Standard deviation	Sharpe ratio
VTSNX	N/A	N/A	0.00	11.58	0.71
Primary benchmark	0.99	1.01	N/A	11.42	0.72
Broad-based benchmark	0.99	1.01	N/A	11.42	0.72

R-squared and beta are calculated from trailing 36-month fund returns relative to the associated benchmark.
Broad-based benchmark: FTSE Global All Cap ex US Index.

Market allocation

	% of common stock	
	VTSNX	Benchmark
Japan	16.7%	16.8%
United Kingdom	11.1	11.3
China	8.3	8.4
Canada	6.6	6.7
France	6.6	6.6
Switzerland	5.7	5.8
Germany	5.5	5.6
Australia	4.7	4.7
Taiwan	3.5	3.5
Korea	3.2	3.2
Other	28.1	27.4

Ten largest holdings

	% of total net assets
Alibaba Group Holding Ltd.	1.4%
Nestle SA	1.2
Tencent Holdings Ltd.	1.1
Taiwan Semiconductor Manufacturing Co. Ltd.	1.0
Samsung Electronics Co. Ltd.	1.0
Royal Dutch Shell plc	0.9
Roche Holding AG	0.9
Novartis AG	0.8
Toyota Motor Corp.	0.7
HSBC Holdings plc	0.6
Top ten as a % of total net assets	**9.6%**

The holdings listed exclude any temporary cash investments and equity index products.

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Vanguard Total Bond Market Index Fund (VBTIX)

Fixed income characteristics

	VBTIX	Benchmark
Number of bonds	9,033	10,982
Average duration	6.2 years	6.0 years
Average effective maturity	8.3 years	8.2 years
Turnover (fiscal year end)	43.9%	—
Short-term reserves	0.0%	N/A

Distribution by issuer type

	% of total net assets
Treasury/Agency	43.3%
Government Mortgage-Backed	22.3%
Industrial	16.5%
Finance	8.5%
Foreign	4.8%
Commercial Mortgage-Backed	2.2%
Utilities	2.0%
Asset-Backed	0.4%

Hedged non-U.S. dollar-denominated bonds are included in the sector of issuer, not as part of the foreign category.

Risk and volatility

	R-squared	Beta	Alpha	Standard deviation	Sharpe ratio
VBTIX	N/A	N/A	-0.00	2.94	0.81
Primary benchmark	0.99	0.99	N/A	2.97	0.82
Broad-based benchmark	0.99	0.99	N/A	2.97	0.82

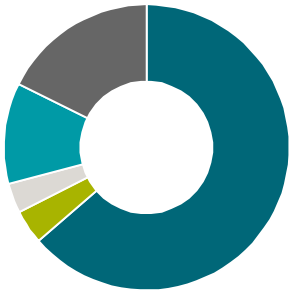
R-squared and beta are calculated from trailing 36-month fund returns relative to the associated benchmark.
Broad-based benchmark: Spliced Bloomberg Barclays U.S. Aggregate Float Adjusted Index.

Distribution by effective maturity as a % of funds



Under 1 Year	0.7%
1 - 3 Years	23.5
3 - 5 Years	25.2
5 - 10 Years	31.7
10 - 20 Years	4.3
20 - 30 Years	14.1
Over 30 Years	0.5

Distribution by credit quality as a % of funds



U.S. Government	63.7%
Aaa	3.8
Aa	3.4
A	11.4
Baa	17.7
Less Than Baa	0.0

Credit-quality ratings are measured on a scale that generally ranges from AAA (highest) to D (lowest). "NR" is used to classify securities for which a rating is not available. NR securities may include a fund's investment in Vanguard Market Liquidity Fund or Vanguard Municipal Cash Management Fund, each of which invests in high-quality money market instruments and may serve as a cash management vehicle for the Vanguard funds, trusts, and accounts. U.S. Treasury, U.S. Agency, and U.S. Agency mortgage-backed securities appear under "U.S. Government." Credit-quality ratings for each issue are obtained from Barclays using ratings derived from Moody's Investors Service (Moody's), Fitch Ratings (Fitch), and Standard & Poor's (S&P). When ratings from all three agencies are available, the median rating is used. When ratings are available from two of the agencies, the lower rating is used. When one rating is available, that rating is used.

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Vanguard Intermediate-Term Bond Index Fund (VBILX)

Fixed income characteristics

	VBILX	Benchmark
Number of bonds	1,875	2,018
Average duration	6.3 years	6.3 years
Average effective maturity	7.2 years	7.1 years
Turnover (fiscal year end)	53.0%	—
Short-term reserves	0.0%	N/A

Distribution by issuer type

	% of total net assets
Treasury/Agency	54.4%
Industrial	24.0%
Finance	14.6%
Foreign	4.7%
Utilities	2.2%
Other	0.1%
Asset-Backed	0.0%
Commercial Mortgage-Backed	0.0%
Government Mortgage-Backed	0.0%

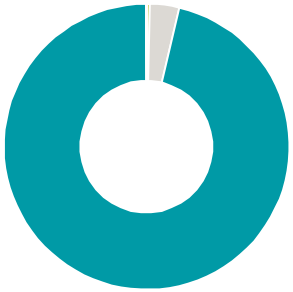
Hedged non-U.S. dollar-denominated bonds are included in the sector of issuer, not as part of the foreign category.

Risk and volatility

	R-squared	Beta	Alpha	Standard deviation	Sharpe ratio
VBILX	N/A	N/A	-0.00	3.41	0.84
Primary benchmark	0.99	0.99	N/A	3.45	0.86
Broad-based benchmark	0.97	1.13	N/A	2.97	0.82

R-squared and beta are calculated from trailing 36-month fund returns relative to the associated benchmark.
Broad-based benchmark: Spliced Bloomberg Barclays U.S. Aggregate Float Adjusted Index.

Distribution by effective maturity as a % of funds



Under 1 Year	0.1%
1 - 3 Years	0.3
3 - 5 Years	3.3
5 - 10 Years	96.2
10 - 20 Years	0.1
20 - 30 Years	0.0
Over 30 Years	0.0

Distribution by credit quality as a % of funds



U.S. Government	53.9%
Aaa	2.1
Aa	3.0
A	16.3
Baa	24.7
Less Than Baa	0.0

Credit-quality ratings are measured on a scale that generally ranges from AAA (highest) to D (lowest). "NR" is used to classify securities for which a rating is not available. NR securities may include a fund's investment in Vanguard Market Liquidity Fund or Vanguard Municipal Cash Management Fund, each of which invests in high-quality money market instruments and may serve as a cash management vehicle for the Vanguard funds, trusts, and accounts. U.S. Treasury, U.S. Agency, and U.S. Agency mortgage-backed securities appear under "U.S. Government." Credit-quality ratings for each issue are obtained from Barclays using ratings derived from Moody's Investors Service (Moody's), Fitch Ratings (Fitch), and Standard & Poor's (S&P). When ratings from all three agencies are available, the median rating is used. When ratings are available from two of the agencies, the lower rating is used. When one rating is available, that rating is used.

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Vanguard Real Estate Index Fund (VGSNX)

Equity characteristics

	VGSNX	Benchmark
Number of stocks	185	181
Median market cap	$18.3B	$17.1B
P/E ratio (trailing earnings)	34.7x	34.7x
P/B ratio	2.7x	2.7x
Return on equity (5-year average)	7.0%	7.0%
Earnings growth rate (5-year)	16.8%	16.8%
Equity yield (dividend)	3.4%	3.4%
Foreign holdings	0.0%	—
Turnover (fiscal year end)	24.0%	—
Short-term reserves	0.0%	N/A

Subindustry diversification as a % of common stock



Sector	VGSNX	Benchmark
Specialized REITs	33.1%	33.2%
Residential REITs	14.6	14.6
Retail REITs	12.0	12.1
Health Care REITs	9.6	9.6
Office REITs	9.6	9.5
Industrial REITs	8.3	8.3
Diversified REITs	4.8	4.8
Hotel & Resort REITs	4.2	4.2
Real Estate Services	2.9	2.9
Real Estate Development	0.4	0.4
Real Estate Operating Companies	0.3	0.3

Sector categories are based on the Global Industry Classification Standard ("GICS"), except for the "Other" category (if applicable), which includes securities that have not been provided a GICS classification as of the effective reporting period.

Market-cap breakdown



Market capitalization	VGSNX	Benchmark
More than $31	29.8%	29.7%
$25 to $31	8.8	8.7
$9 to $25	31.7	31.9
$5 to $9	13.7	13.7
Below $5	16.0	16.0

Risk and volatility

	R-squared	Beta	Alpha	Standard deviation	Sharpe ratio
VGSNX	N/A	N/A	-0.01	12.07	0.56
Primary benchmark	1.00	1.00	N/A	12.07	0.56
Broad-based benchmark	0.29	0.53	N/A	12.39	1.04

R-squared and beta are calculated from trailing 36-month fund returns relative to the associated benchmark.
Broad-based benchmark: Dow Jones U.S. Total Stock Market Float Adjusted Index.

Ten largest holdings

	% of total net assets
Vanguard Real Estate II Index Fund	11.1%
American Tower Corp.	7.1
Crown Castle International Corp.	4.1
Prologis Inc.	3.9
Equinix Inc.	3.5
Simon Property Group Inc.	3.2
Public Storage	2.3
Welltower Inc.	2.3
Equity Residential	2.1
AvalonBay Communities Inc.	2.0
Top ten as a % of total net assets	**41.6%**

The holdings listed exclude any temporary cash investments and equity index products.

All data as of December 31, 2019, unless otherwise noted.

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Important information

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The VCMM projections are based on a statistical analysis of historical data. Future returns may behave differently from the historical patterns captured in the VCMM. More importantly, the VCMM may be underestimating extreme negative scenarios unobserved in the historical period on which the model estimation is based.

The Vanguard Capital Markets Model® is a proprietary financial simulation tool developed and maintained by Vanguard's primary investment research and advice teams. The model forecasts distributions of future returns for a wide array of broad asset classes. Those asset classes include U.S. and international equity markets, several maturities of the U.S. Treasury and corporate fixed income markets, international fixed income markets, U.S. money markets, commodities, and certain alternative investment strategies. The theoretical and empirical foundation for the Vanguard Capital Markets Model is that the returns of various asset classes reflect the compensation investors require for bearing different types of systematic risk (beta). At the core of the model are estimates of the dynamic statistical relationship between risk factors and asset returns, obtained from statistical analysis based on available monthly financial and economic data from as early as 1960. Using a system of estimated equations, the model then applies a Monte Carlo simulation method to project the estimated interrelationships among risk factors and asset classes as well as uncertainty and randomness over time. The model generates a large set of simulated outcomes for each asset class over several time horizons. Forecasts are obtained by computing measures of central tendency in these simulations. Results produced by the tool will vary with each use and over time.

The Vanguard Lifecycle Model (VLCM) is designed to identify the product design that represents the best investment solution for a theoretical, representative investor who uses the target-date funds to accumulate wealth for retirement. The VLCM generates an optimal custom glide path for a participant population by assessing the trade-offs between the expected (median) wealth accumulation and the uncertainty about that wealth outcome, for thousands of potential glide paths. The VLCM does this by combining two set of inputs: the asset class return projections from the VCMM and the average characteristics of the participant population. Along with the optimal custom glide path, the VLCM generates a wide range of portfolio metrics such as a distribution of potential wealth accumulation outcomes, risk and return distributions for the asset allocation, and probability of ruin, such as the odds of participants depleting their wealth by age 95.

The VLCM inherits the distributional forecasting framework of the VCMM and applies to it the calculation of wealth outcomes from any given portfolio.

The most impactful drivers of glide path changes within the VLCM tend to be risk aversion, the presence of a defined benefit plan, retirement age, savings rate and starting compensation. The VLCM chooses among glide paths by scoring them according to the utility function described and choosing the one with the highest score. The VLCM does not optimize the levels of spending and contribution rates. Rather, the VLCM optimizes the glide path for a given customizable level of spending, growth rate of contributions and other plan sponsor characteristics.

A full dynamic stochastic life-cycle model, including optimization of a savings strategy and dynamic spending in retirement is beyond the scope of this framework.

Important information

For more information about any fund, visit institutional.vanguard.com or call 866-499-8473 to obtain a prospectus or, if available, a summary prospectus. Investment objectives, risks, charges, expenses, and other important information are contained in the prospectus; read and consider it carefully before investing.

Vanguard ETF® Shares are not redeemable with the issuing fund other than in very large aggregations worth millions of dollars. Instead, investors must buy or sell Vanguard ETF Shares in the secondary market and hold those shares in a brokerage account. In doing so, the investor may incur brokerage commissions and may pay more than net asset value when buying and receive less than net asset value when selling.

Mutual funds and all investments are subject to risk, including the possible loss of the money you invest. Prices of mid- and small-cap stocks often fluctuate more than those of large-company stocks. Investments in stocks or bonds issued by non-U.S. companies are subject to risks including country/regional risk and currency risk. These risks are especially high in emerging markets. Funds that concentrate on a relatively narrow sector face the risk of higher share-price volatility. It is possible that tax-managed funds will not meet their objective of being tax-efficient. Because company stock funds concentrate on a single stock, they are considered riskier than diversified stock funds.

Investments in bond funds are subject to the risk that an issuer will fail to make payments on time and that bond prices will decline because of rising interest rates or negative perceptions of an issuer's ability to make payments. High-yield bonds generally have medium- and lower-range credit quality ratings and are therefore subject to a higher level of credit risk than bonds with higher credit quality ratings. Although the income from a municipal bond fund is exempt from federal tax, you may owe taxes on any capital gains realized through the fund's trading or through your own redemption of shares. For some investors, a portion of the fund's income may be subject to state and local taxes, as well as to the federal Alternative Minimum Tax. Diversification does not ensure a profit or protect against a loss.

Investments in Target Retirement Funds or Trusts are subject to the risks of their underlying funds. The year in the fund name refers to the approximate year (the target date) when an investor in the fund or trust would retire and leave the workforce. The fund or trust will gradually shift its emphasis from more aggressive investments to more conservative ones based on its target date. An investment in the Target Retirement Fund or Trust is not guaranteed at any time, including on or after the target date.

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Vanguard Marketing Corporation, Distributor of the Vanguard Funds. U.S. Patent Nos. 6,879,964; 7,337,138; 7,720,749; 7,925,573; 8,090,646; 8,417,623; and 8,626,636.

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